Power Budget

Team Number:	105
Project Name:	**Speaker Board**
Team Member Names:	**William Buckner**
Version:	**Rev 3 - 5 V single-rail design**

+5V Power Rail	**Component Name**	**Part Number**	**Supply Voltage Range**
	PIC18F57Q43 Curiosity Nano	DM164150	1.8-5.5 V
	10-bit DAC	MCP4911	2.7-5.5 V
	Quad op amp	MCP6004	1.8-6.0 V
	Class-AB speaker output	BD139 + BD140	5 V design
	Debug / power LEDs	D1 + D4 + D5	5 V via resistors
	Subtotal		
	Safety Margin		
	Total Current Required on +5V Rail		
c2. Regulator or Source Choice	+5V Linear Regulator	LM7805	7-35 V
	Total Remaining Current Available on +5V Rail		

External Power Source 1	**Component Name**	**Part Number**	**Supply Voltage Range**
Power Source 1 Selection	9 V DC Wall Supply	TBD - >=1 A	120 VAC
Power Rails Connected to External Power Source 1	+5V Linear Regulator	LM7805	7-35 V

Total Remaining Current Available on External Power Source 1
Notes
BD139/BD140 row uses 300 mA estimated worst-case 5 V rail draw for the complete Class-AB output stage. 5 V rail total includes 25% safety margin; verify regulator temperature and measured current during hardware testing.

#	Absolute Maximum Current (mA)	Total Current (mA)	Unit
1	100	100	mA
1	0.55	0.55	mA
1	0.68	0.68	mA
1	300	300	mA
3	5	15	mA
		416.23	mA
		25%	
		520.2875	mA
1	1000	1000	mA
		479.7125	mA

Output Voltage	Absolute Maximum Current (mA)	Total Current (mA)	Unit
9 VDC	1000	1000	mA
1	1000	520.2875	mA
			mA
		1000	mA

	-520.2875	mA